HDOC Day 43

Aim

To write a menu-driven C program using separate functions to find the longest word in a sentence and check whether one string is a rotation of another.

Algorithm

1. Start the program and display the menu.

2. For option 1, read a sentence, scan its words and track the longest word.

3. For option 2, read two strings and check that their lengths match.

4. Concatenate the first string with itself; search for the second string in the concatenated string.

5. Display the result and repeat the menu until Exit is selected.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
#define MAX 500

void readLine(char str[], int size) {
    if (fgets(str, size, stdin) != NULL)
        str[strcspn(str, "\n")] = '\0';
}

void findLongestWord(const char sentence[], char longest[]) {
    int i = 0, start = 0, bestStart = 0;
    int currentLen = 0, bestLen = 0;
    while (1) {
        unsigned char ch = (unsigned char)sentence[i];
        if (ch != '\0' && !isspace(ch)) {
            if (currentLen == 0) start = i;
            currentLen++;
        } else {
            if (currentLen > bestLen) {
                bestLen = currentLen;
                bestStart = start;
            }
            currentLen = 0;
            if (ch == '\0') break;
        }
        i++;
    }
    strncpy(longest, sentence + bestStart, bestLen);
    longest[bestLen] = '\0';
}

int isRotation(const char first[], const char second[]) {
    char doubled[2 * MAX];
    if (strlen(first) != strlen(second)) return 0;
    strcpy(doubled, first);
    strcat(doubled, first);
    return strstr(doubled, second) != NULL;
}

int main(void) {
    char sentence[MAX], first[MAX], second[MAX];
```

```
    char longest[MAX];
    int choice;
    while (1) {
        printf("\n--- MENU ---\n");
        printf("1. Find longest word\n");
        printf("2. Check string rotation\n");
        printf("3. Exit\nEnter choice: ");
        if (scanf("%d", &choice) != 1) {
            printf("Invalid input.\n");
            break;
        }
        while (getchar() != '\n') { }
        switch (choice) {
            case 1:
                printf("Enter a sentence: ");
                readLine(sentence, MAX);
                findLongestWord(sentence, longest);
                if (longest[0] == '\0')
                    printf("No words found.\n");
                else
                    printf("Longest word: %s\n", longest);
                break;
            case 2:
                printf("Enter first string: ");
                readLine(first, MAX);
                printf("Enter second string: ");
                readLine(second, MAX);
                if (isRotation(first, second))
                    printf("Strings are rotations.\n");
                else
                    printf("Strings are not rotations.\n");
                break;
            case 3:
                printf("Exiting program.\n");
                return 0;
            default:
                printf("Invalid choice. Try again.\n");
        }
    }
    return 0;
}
```

Sample Output

```
--- MENU ---
1. Find longest word
2. Check string rotation
3. Exit
Enter choice: 1
Enter a sentence: Programming is interesting
Longest word: Programming

Enter choice: 2
Enter first string: abcde
Enter second string: cdeab
Strings are rotations.

Enter choice: 3
Exiting program.
```

Result

The menu-driven C program was successfully implemented to find the longest word in a sentence and determine whether two strings are rotations of each other.